

1.1 INTRODUCTION

Traditional PR models assume **crisp (hard) decision boundaries**.

But real-world data is:

- Noisy
- Overlapping
- Vague
- Imprecise
- Ambiguous

Hence, **Fuzzy Logic (FL)** becomes crucial.

1.2 Why Fuzzy Logic

What is Fuzzy Logic?

Fuzzy Logic (FL) is a method of reasoning that resembles **human reasoning**. This approach is similar to how humans perform decision making. And it involves all intermediate possibilities between **YES** and **NO**.

Fuzzy Logic models:

- Ambiguity
- Uncertainty
- Vagueness
- Partial truth
- Human reasoning

Example in real-world:

A leaf may be:

- **0.6 diseased**
- **0.4 healthy**

A pixel may be:

- **0.7 tumor-like**
- **0.3 non-tumor**

Traditional classifiers cannot model such continuous ambiguity. Fuzzy classifiers can.

1.3 Why do we use Fuzzy Logic?

Generally, we use the fuzzy logic system for both commercial and practical purposes such as:

- It **controls machines** and **consumer products**

- If not accurate reasoning, it at least provides **acceptable reasoning**
- This helps in dealing with the **uncertainty in engineering**

1.4. Real-world applications of fuzzy logic

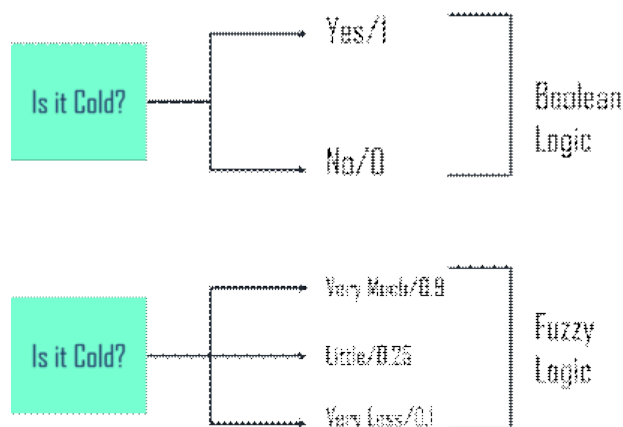
Fuzzy logic is behind many of our daily technologies:

- **Home appliances:** Optimizing performance in washing machines and air conditioners
- **Automotive systems:** Enhancing cruise control and gear shifting
- **Medical diagnosis:** Evaluating symptoms with varying severity

1.5 Core Concepts

• Fuzzy logic vs binary logic

- **Binary logic:** Binary is 0 or 1, or TRUE or FALSE .
- **Fuzzy logic:** Allows for intermediate states (0.8,0.2), such as very much or little, for more realistic decision-making.



• Linguistic Variables:

- These are variables defined using terms rather than precise numbers.
- For example, "temperature" could be a linguistic variable with terms like "cold," "warm" and "hot."

• Fuzzy sets :

- Proposed by Lotfi A. Zadeh (1965).
- These are sets where elements have degrees of membership. For example, the "young" age group might include someone who is 30 with 0.7 membership (mostly young) and someone who is 50 with 0.3 membership (partly young).

A fuzzy set A is defined as:

$$A = \{(x, \mu_A(x)) | x \in X\}$$

Where:

- x = element
- $\mu_A(x)$ = membership degree

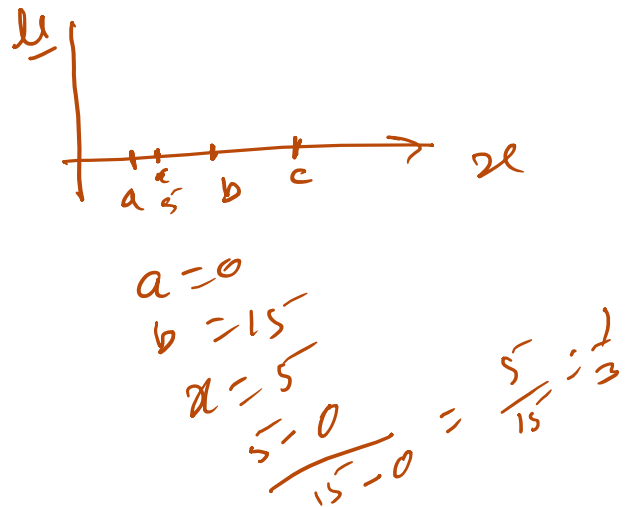
• Membership Functions:

- Define how much a value belongs to a fuzzy set
- Example- 30°C is 0.7 "Warm" and 0.3 "Hot".

• Membership Function (MF) Types

1. Triangular MF

$$\mu(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ \frac{c-x}{c-b} & b \leq x \leq c \\ 0 & x \geq c \end{cases}$$



2. Trapezoidal MF

Piecewise linear.

3. Gaussian MF

$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right)$$

Smooth, widely used in neuro-fuzzy models.

4. Bell-shaped MF

$$\mu(x) = \frac{1}{1 + \left|\frac{x-c}{a}\right|^{2b}}$$

5. Sigmoidal MF

$$\mu(x) = \frac{1}{1 + e^{-a(x-c)}}$$

Fuzzy rules and inference systems

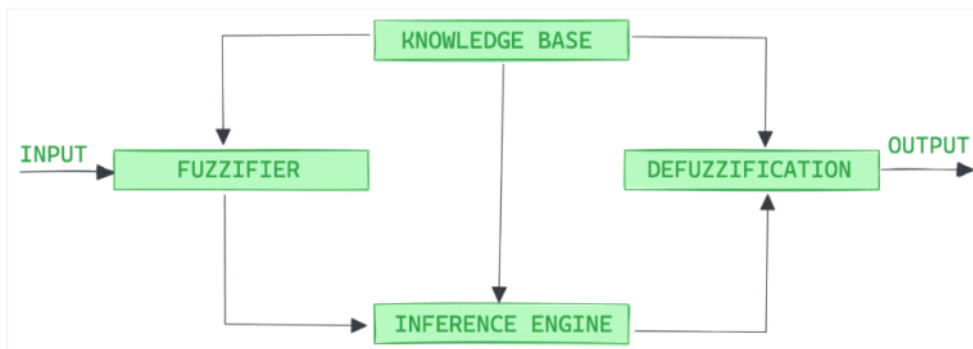
- Fuzzy logic operates through IF-THEN rules, which express relationships between inputs and outputs in a fuzzy way. For instance:
- IF temperature is "moderately hot" THEN fan speed is "medium."

These rules are processed by fuzzy inference systems, which combine multiple fuzzy rules to make decisions. The process involves:

- Evaluating the degree of truth for each rule based on the input values.
- Combining the results of all applicable rules using logical operations.
- Generating a fuzzy output, which is then converted into a precise value using defuzzification methods.

1.6 Fuzzy Logic Architecture

The fuzzy logic architecture consists of four main parts:



1. Fuzzification

- Fuzzification converts crisp inputs—precise values measured by sensors, such as temperature or pressure—into fuzzy values using membership functions.
- These fuzzy values represent degrees of membership in different categories (e.g., "cold," "warm," or "hot"), enabling the system to handle imprecision effectively.

2. Knowledge base

- The knowledge base contains a set of IF-THEN rules and membership functions provided by experts.
- These rules govern the decision-making process by translating linguistic inputs into actionable outputs.
- Advances in fuzzy logic design have streamlined the development and tuning of fuzzy controllers, often reducing the number of rules required for effective operation.

3. Inference engine

- The inference engine emulates human reasoning by evaluating the degree of match between fuzzy inputs and the rules stored in the knowledge base.
- Based on this evaluation, it determines which rules to activate and combines their outcomes to form control actions.
- This process is the core of decision-making in a fuzzy logic system.

4. Defuzzification

- Defuzzification transforms the fuzzy output generated by the inference engine into a crisp value.
- This step translates the fuzzy control actions into precise, actionable outputs suitable for real-world applications.
- Several defuzzification techniques exist, and the choice of method depends on the specific system requirements.

Fuzzy Logic Demo Example: Temperature Control System

Goal:

Control the **fan speed** based on **room temperature** using fuzzy logic.

Problem:

We want to control **fan speed** based on **room temperature**.

Input:

Room Temperature (0–50°C)

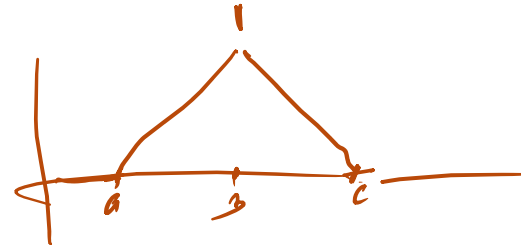
Output:

Fan Speed (0–100%)

1. Define the Membership Functions Clearly

$$\mu(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ \frac{c-x}{c-b}, & b < x < c \\ 0, & x \geq c \end{cases}$$

30 ≤ 36



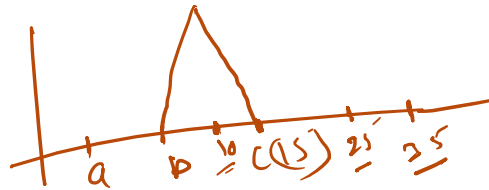
1.1 Temperature Fuzzy Sets

Cold

Triangular MF: (0, 0, 15)

Meaning:

- Fully Cold at 0
- Starts decreasing after 0
- Becomes not cold at 15

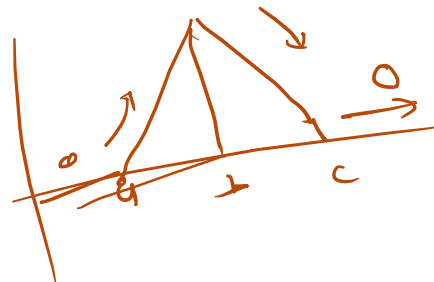


Warm

Triangular MF: (10, 25, 35)

Hot

Triangular MF: (30, 50, 50)



1.2 Fan Speed Fuzzy Sets

Slow

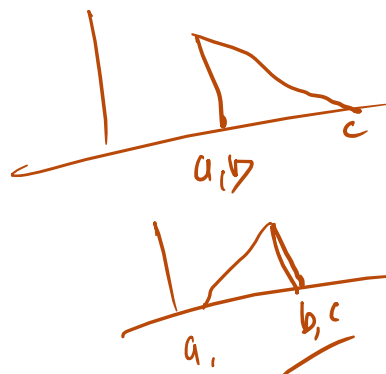
(0, 0, 40)

Moderate

(30, 50, 70)

Fast

(60, 100, 100)



2. Fuzzy Rules

1. IF Temperature is Cold → Fan Speed = Slow
2. IF Temperature is Warm → Fan Speed = Moderate
3. IF Temperature is Hot → Fan Speed = Fast

3. Now Take an Example Input

Let the temperature = 30°C

Now we check the membership values.

4. How to Calculate Membership Values

4.1 Membership of 30°C in Cold

Cold = (0, 0, 15)

4.1 Membership of 30°C in Cold

Cold = (0,0,15)

Since 30 is outside the range,

Cold(30) = 0

$$\begin{array}{l} 27/15 \rightarrow 0 \\ 30/15 \rightarrow 0 \end{array}$$

4.2 Membership of 30°C in Warm

Warm = (10, 25, 35)

This is a triangle:

^

/ \

10 25 35

For 30°C, use the **descending slope equation**:

$$\mu = (35 - 30) / (35 - 25)$$

$$\mu = (5) / (10) = 0.5$$

So:

Warm(30) = 0.5

$$\mu = \frac{c-x}{c-b} = \frac{35-30}{35-25} = \frac{5}{10} = 0.5$$

4.3 Membership of 30°C in Hot

Hot = (30, 50, 50)

Shape:

30 50

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Since 30 is the left base point:

Hot(30) = 0 (triangular MF starts rising after 30)

To compute properly:

For x = 30,

$$\text{Hot} = (x - 30) / (50 - 30) = (0) / 20 = 0$$

Summary of Membership Values

Temperature Fuzzy Set μ value

Cold 0

Warm 0.5

Hot 0

Only **Warm** is active.

5. Apply the Rule

Rule triggered:

IF Temperature is Warm $\rightarrow$ Fan Speed = Moderate

Strength = 0.5

This means the output fuzzy set **Moderate** is cut at height 0.5.

6. Defuzzification (Center of Gravity)

Moderate MF = (30, 50, 70)

We scale it down to 0.5 height.

The centroid of a symmetric triangle is its center = **50**

When scaled (clipped), the centroid remains 50.

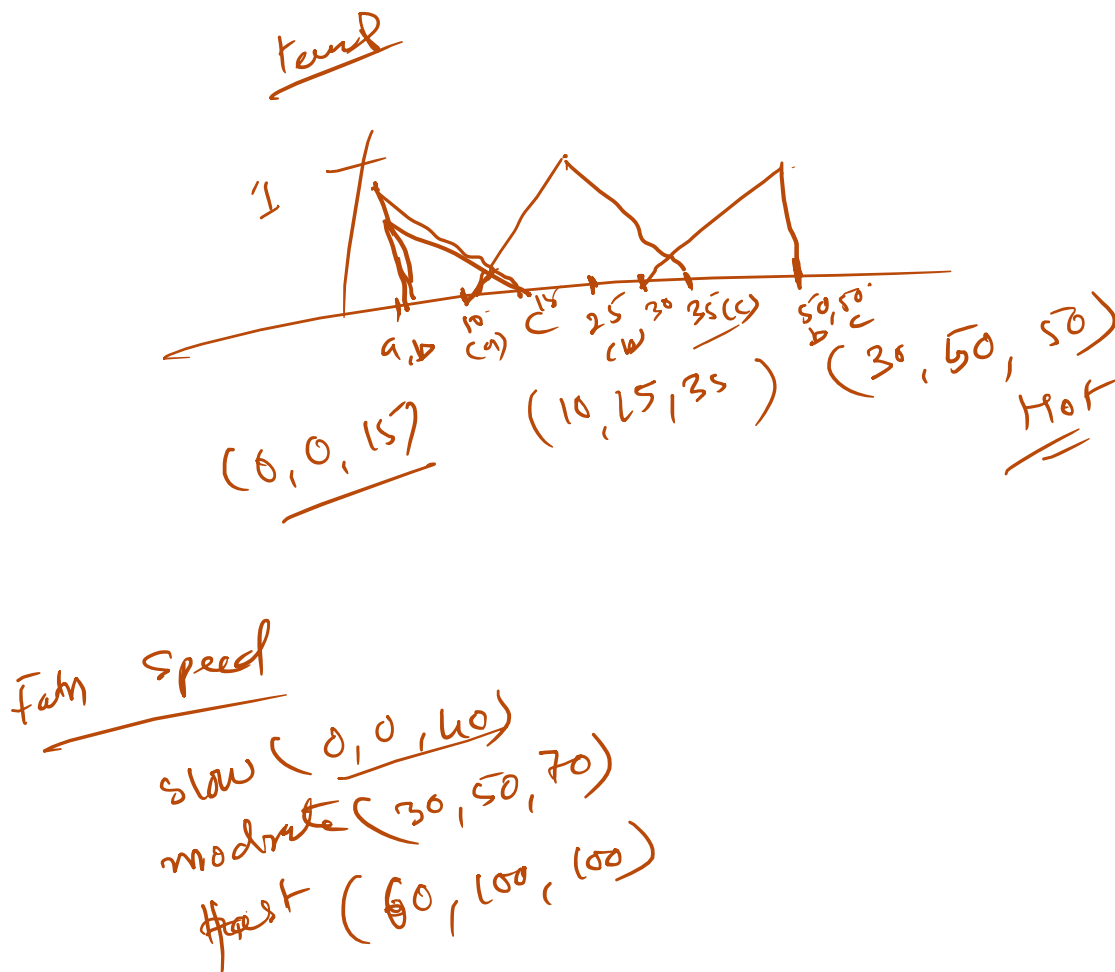
So final output fan speed = **50%**

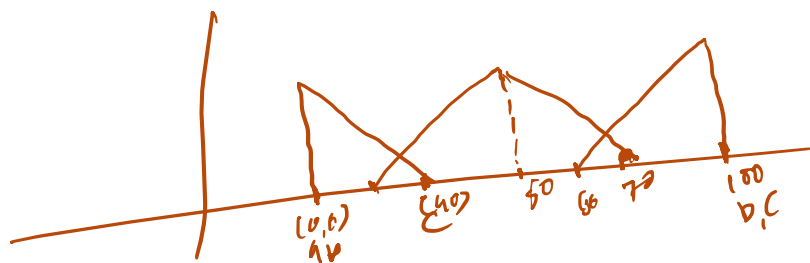
Final Answer

If temperature = **30°C**, fan speed = **50%**

Would you like:

1. A graphical explanation with ASCII plots?
2. A Python implementation?
3. A MATLAB example?
4. A fuzzy control system diagram?



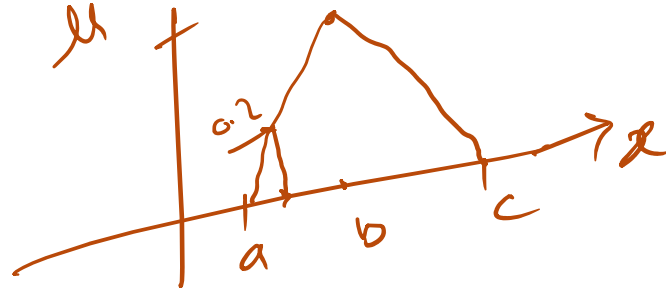
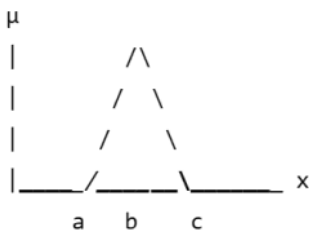


What Does a Triangular Membership Function (a, b, c) Mean?

A triangular membership function has 3 points:

- **a = left base point** (membership = 0)
- **b = peak point** (membership = 1)
- **c = right base point** (membership = 0)

Its shape:

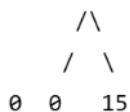


Now Understand This Example: (0, 0, 15)

Here:

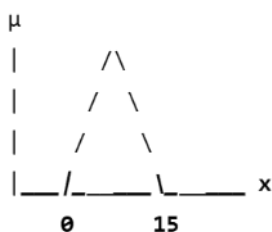
- $a = 0$
- $b = 0$
- $c = 15$

So the triangle becomes:



But because $a = b$, the triangle is not slanted on the left side — it rises vertically from 0.

Let me draw it properly:



At $x = 0$, membership = 1 (peak).

At $x = 15$, membership = 0.

Why Do We Use (0, 0, 15)?

This defines the fuzzy set **Cold**.

Interpretation:

- Temperature **0°C is fully Cold** $\rightarrow \mu(0) = 1$
- As temperature increases above 0, the “Coldness” decreases
- At 15°C $\rightarrow$ **no longer Cold** $\rightarrow \mu(15) = 0$

So the membership gradually drops from 1 to 0 between 0 and 15.

This is a **left-shoulder triangular membership**.

Mathematically:

For Triangular MF (a, b, c):

$$0, \quad x \leq a \text{ or } x \geq c$$

$$(x-a)/(b-a), \quad a < x \leq b$$

$$(c-x)/(c-b), \quad b < x < c$$

For (0, 0, 15):

- **a = 0**
- **b = 0**
- **c = 15**

Let's plug them in:

Region 1: $x = 0$

Since $x = b \rightarrow \mu(0) = 1$

Region 2: $0 < x < 15$

Use: $(c - x) / (c - b)$

$$= (15 - x) / 15$$

Example:

$$x = 5 \rightarrow \mu = (15 - 5) / 15 = 10/15 = 0.67$$

$$x = 10 \rightarrow \mu = 5/15 = 0.33$$

Region 3: $x \geq 15$

$$\mu = 0$$

Summary in Simple Words

Triangular MF (0, 0, 15) means:

- Value is **fully Cold at 0**
- Coldness decreases linearly as temperature increases
- After 15°C, temperature is **not Cold**

This is how fuzzy logic describes "Cold".